Linear Systems of Equations

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1 A Motivating Example

Over the field $\mathbb{R}$, consider the linear system

$$2x + y = 5, \tag{1a}$$
$$x - y = 1. \tag{1b}$$

Adding the two equations eliminates y:

$$3x = 6. \tag{1a+1b}$$

From (1a+1b), we get $x = 2$, and substituting into (1b) (or (1a)) then gives $y = 1$. The unique solution is therefore

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

2 Linear Systems in General

Let K be a field and $m, n \in \mathbb{N}$. A *linear system* of n equations in m unknowns is an equation $Ax = b$ with coefficient matrix $A \in K^{n \times m}$, right-hand side $b \in K^n$ and variable $x \in K^m$:

$$Ax = \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = b.$$

We seek its *solution set*

$$L(A, b) \coloneqq \{x \in K^m \mid Ax = b\}. \tag{2}$$

The system is solvable if and only if b lies in the column space of A, which is equivalent to $\operatorname{rank}(A) = \operatorname{rank}([A \mid b])$, where $[A \mid b] \in K^{n \times (m+1)}$ is the right hand side augmented coefficient matrix. Provided we find any one particular solution x_0, then $L(A, b) = x_0 + L(A, 0)$, and $L(A, 0)$ is a subspace of K^m of dimension $m - \operatorname{rank}(A)$. Accordingly, the set defined in (2) is either empty or has the same cardinality as the solution set to the homogeneous system. More specifically, we have that

$$|L(A, b)| = \begin{cases} 0 & \text{if } \operatorname{rank}(A) \neq \operatorname{rank}([A \mid b]), \\ |L(A, 0)| = 1 & \text{if } \operatorname{rank}(A) = \operatorname{rank}([A \mid b]) = m, \\ |L(A, 0)| > 1 & \text{if } \operatorname{rank}(A) = \operatorname{rank}([A \mid b]) < m. \end{cases}$$

For the system in (1), we have $n = m = 2$ and $\operatorname{rank}\left(\begin{smallmatrix} 2 & 1 \\ 1 & -1 \end{smallmatrix}\right) = \operatorname{rank}\left(\begin{smallmatrix} 2 & 1 \\ 1 & -1 \end{smallmatrix} \middle| \begin{smallmatrix} 5 \\ 1 \end{smallmatrix}\right) = m$, so it is uniquely solvable, in agreement with the explanation in section 1.